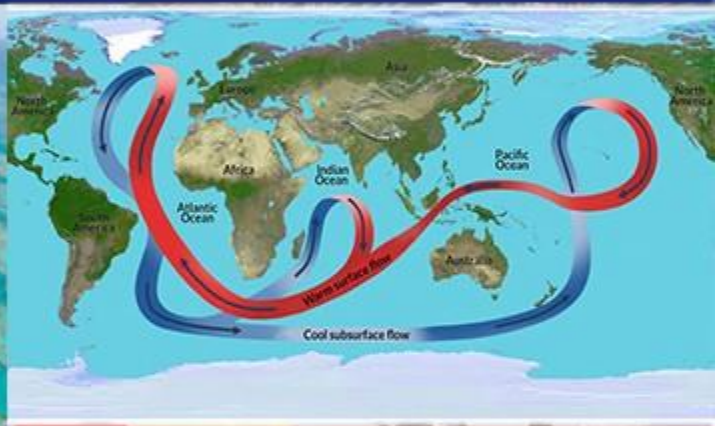


GEOGRAPHY

PHYSICAL GEOGRAPHY



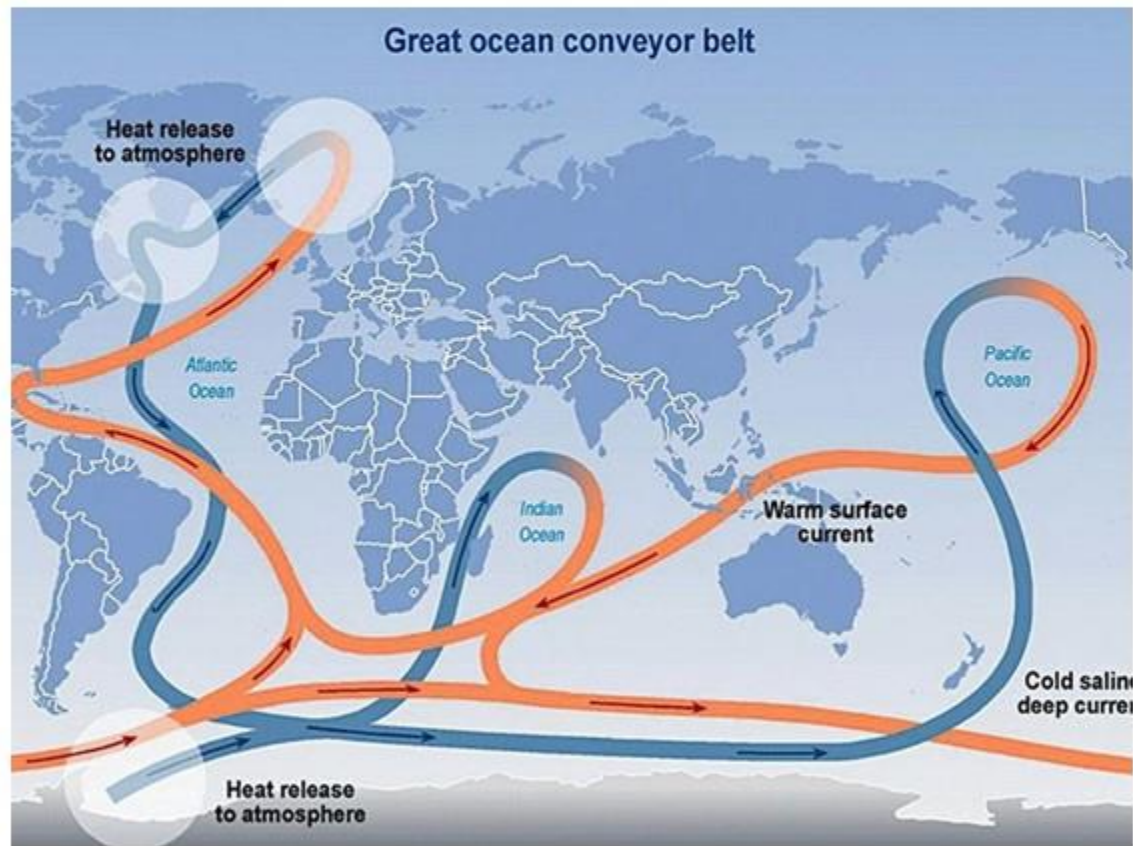
OCEANOGRAPHY

MODULE - 6

OCEAN CURRENTS

MOVEMENTS OF OCEAN WATER

OCEAN CURRENTS

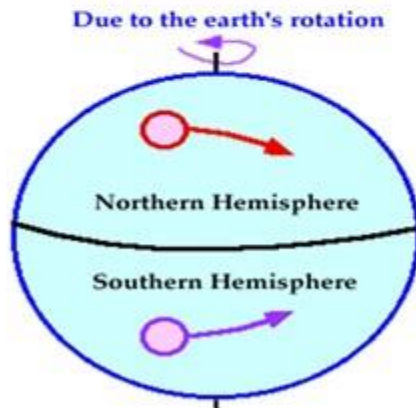


- 1 Ocean currents are like river flow in oceans.
- 2 They represent a regular volume of water in a definite path and direction.
- 3 Ocean currents are influenced by two types of forces namely :
 - i. primary forces that initiate the movement of water
 - ii. secondary forces that influence the currents to flow.

MOVEMENTS OF OCEAN WATER

OCEAN CURRENTS

WHAT ARE THE PRIMARY FORCES THAT INFLUENCE THE CURRENTS?



4 The primary forces that influence the currents are:

- Heating by solar energy
- Wind
- Gravity
- Coriolis force.

5 Heating by solar energy causes the water to expand.

6 That is why, near the equator the ocean water is about 8 cm higher in level than in the middle latitudes.

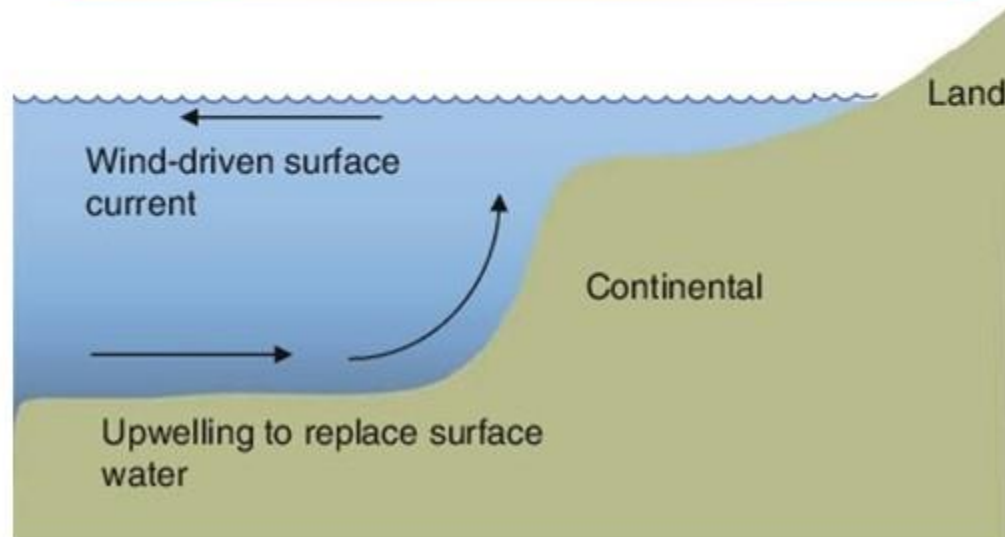
7 This causes a **very slight gradient** and water **tends to flow down the slope**.

MOVEMENTS OF OCEAN WATER

OCEAN CURRENTS

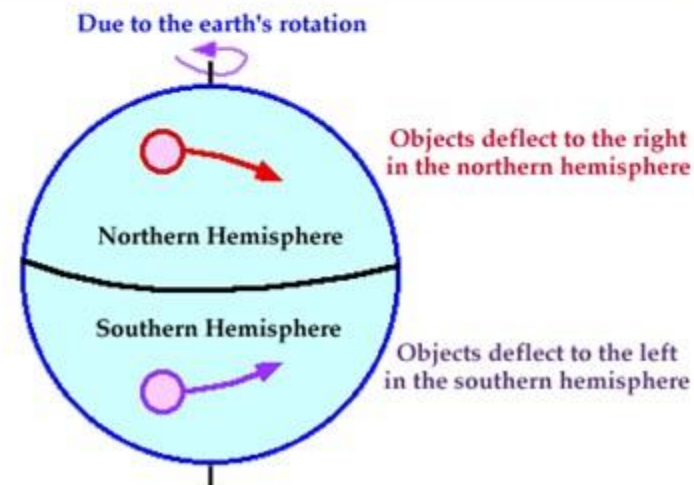
WHAT ARE THE PRIMARY FORCES THAT INFLUENCE THE CURRENTS?

- 8** Wind blowing on the surface of the ocean pushes the water to move.
- 9** Friction between the wind and the water surface affects the movement of the water body in its course.



- 10** Gravity tends to pull the water down the pile and create gradient variation.

- 11** The Coriolis force intervenes and causes the water to move to the right in the northern hemisphere and to the left in the southern hemisphere.

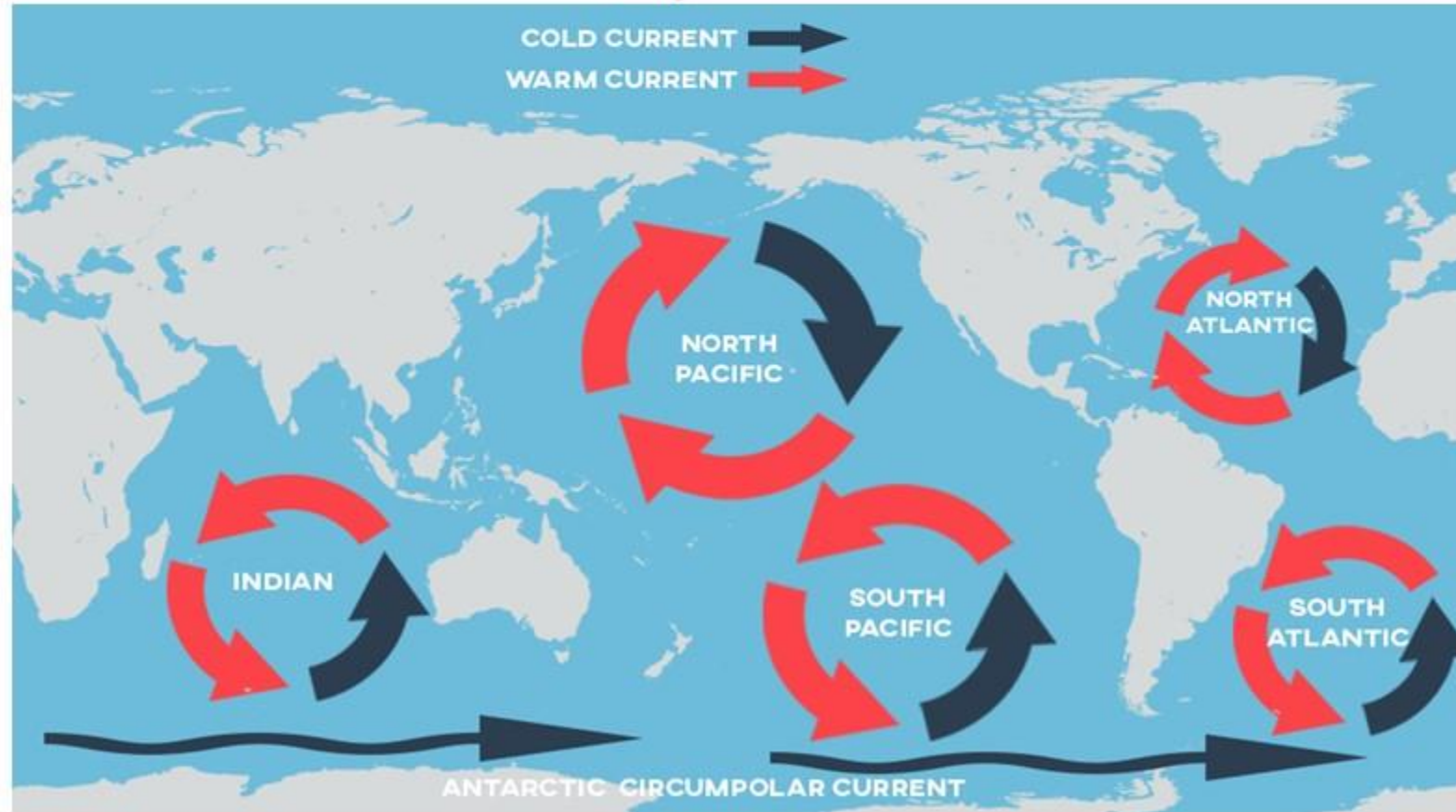


MOVEMENTS OF OCEAN WATER

OCEAN CURRENTS

12 These large accumulations of water and the flow around them are called **Gyres**.

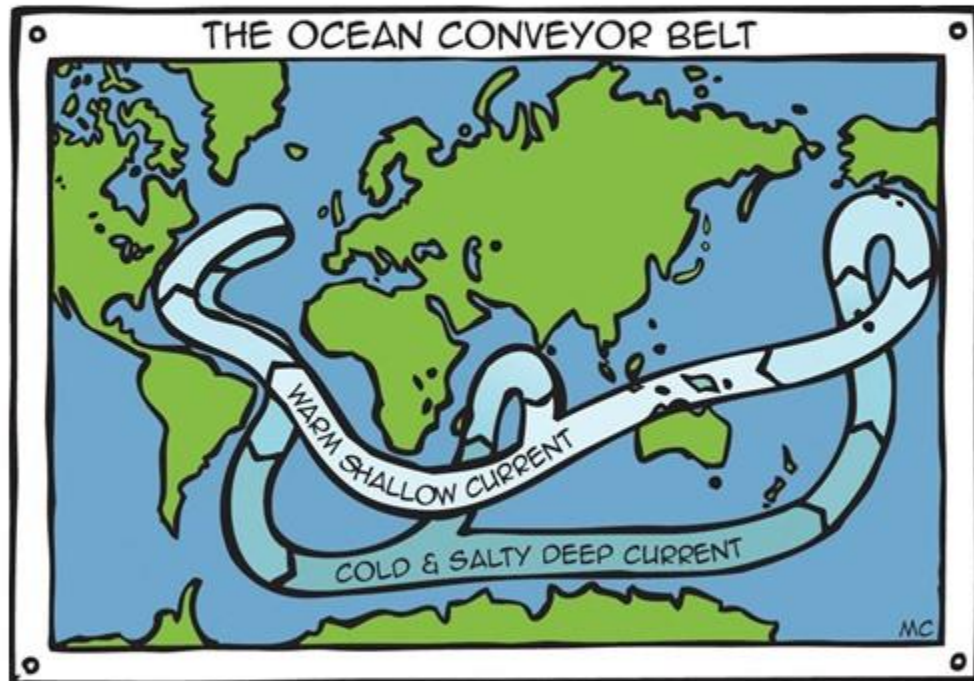
13 These produce large circular currents in all the ocean basins.



MOVEMENTS OF OCEAN WATER

OCEAN CURRENTS

WHAT ARE THE SECONDARY FORCES THAT INFLUENCE THE CURRENTS?



- 14** Differences in water density affect vertical mobility of ocean currents.

- 15** Water with high salinity is denser than water with low salinity and in the same way cold water is denser than warm water.

- 16** Denser water tends to sink, while relatively lighter water tends to rise.

- 17** Cold-water ocean currents occur when the cold water at the poles sinks and slowly moves towards the equator.

- 18** Warm-water currents travel out from the equator along the surface, flowing towards the poles to replace the sinking cold water.

MOVEMENTS OF OCEAN WATER

CHARACTERISTICS OF OCEAN CURRENTS

1 Currents are referred to by their “drift”.

2 Usually, the currents are strongest near the surface and may attain speeds over five knots.

3 At depths, currents are generally slow with speeds less than 0.5 knots.

4 We refer to the speed of a current as its “drift.”

5 Drift is measured in terms of knots.

6 The strength of a current refers to the speed of the current.

7 A fast current is considered strong.

8 A current is usually strongest at the surface and decreases in strength (speed) with depth.

9 Most currents have speeds less than or equal to 5 knots.

MOVEMENTS OF OCEAN WATER

TYPES OF OCEAN CURRENTS

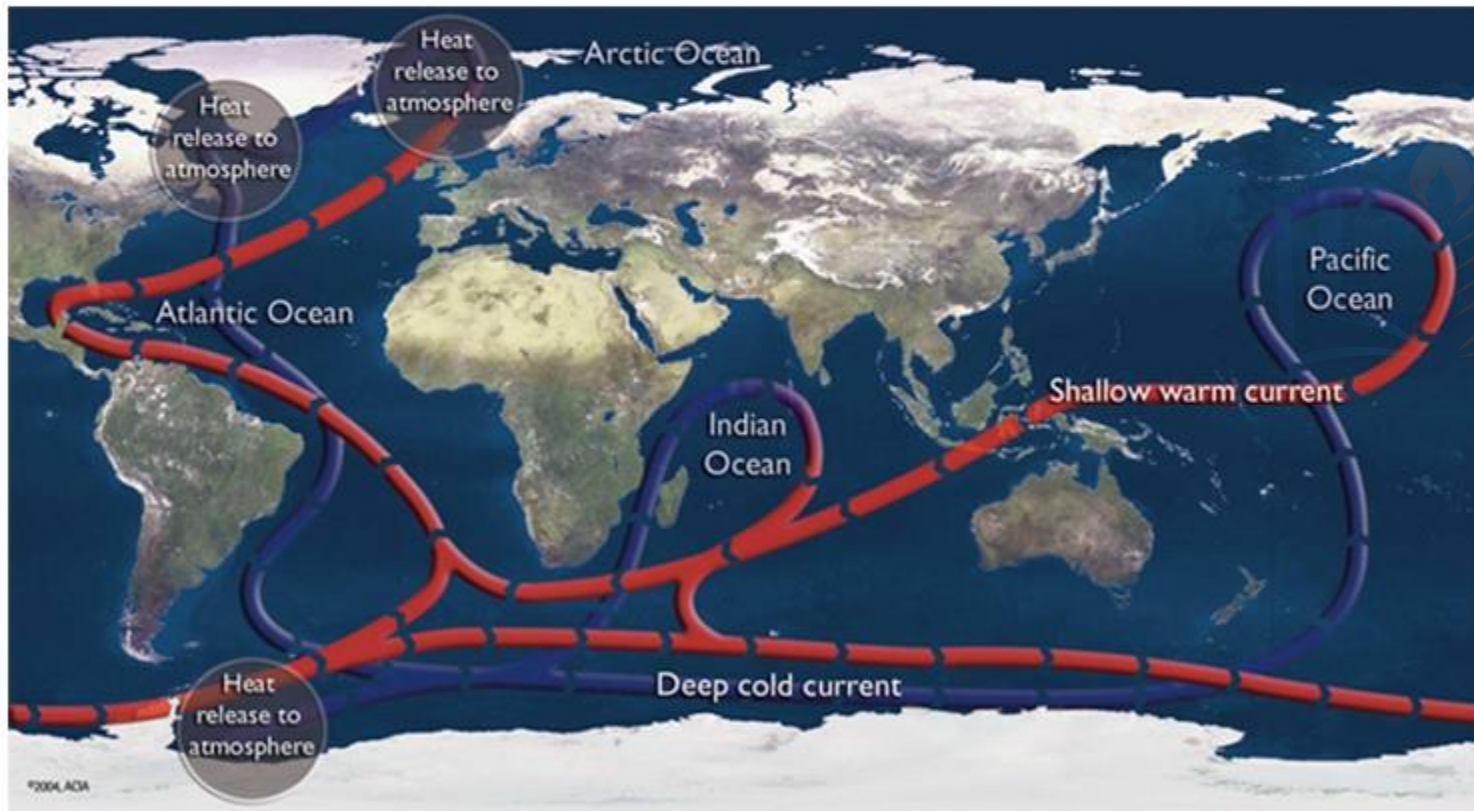
1

CLASSIFICATION OF CURRENTS ON THE BASIS OF DEPTH

1

The ocean currents may be classified based on their depth as **surface currents** and **deep water currents** :

- i. **Surface currents** constitute about 10 per cent of all the water in the ocean, these waters are the upper 400 m of the ocean
- ii. **Deep water currents** make up the other 90 per cent of the **ocean water**. These waters move around the ocean basins due to variations in the density and gravity.



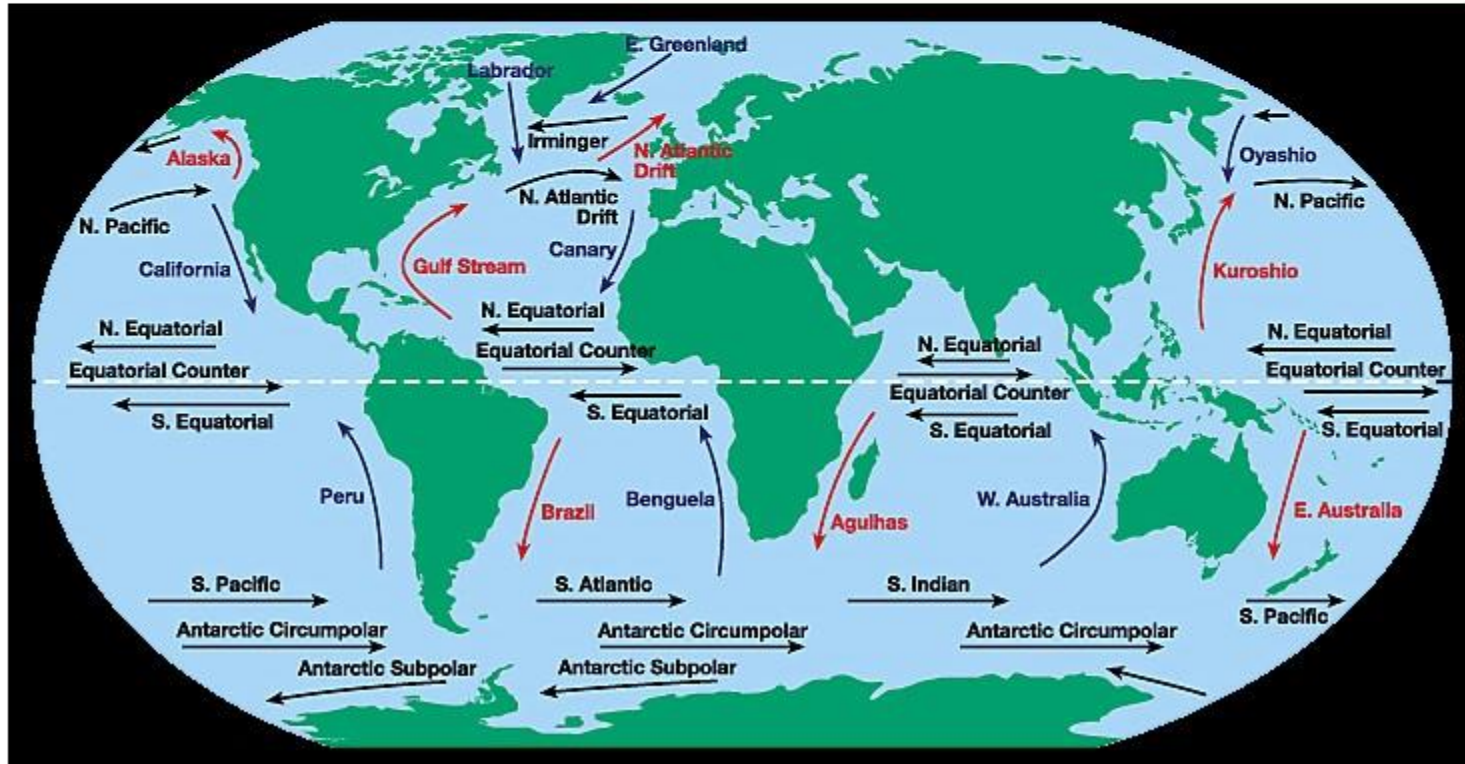
MOVEMENTS OF OCEAN WATER

TYPES OF OCEAN CURRENTS

2

CLASSIFICATION OF CURRENTS ON THE BASIS OF TEMPERATURE

Ocean currents can also be classified based on temperature as cold currents and warm currents:



2

i. **COLD CURRENTS:**

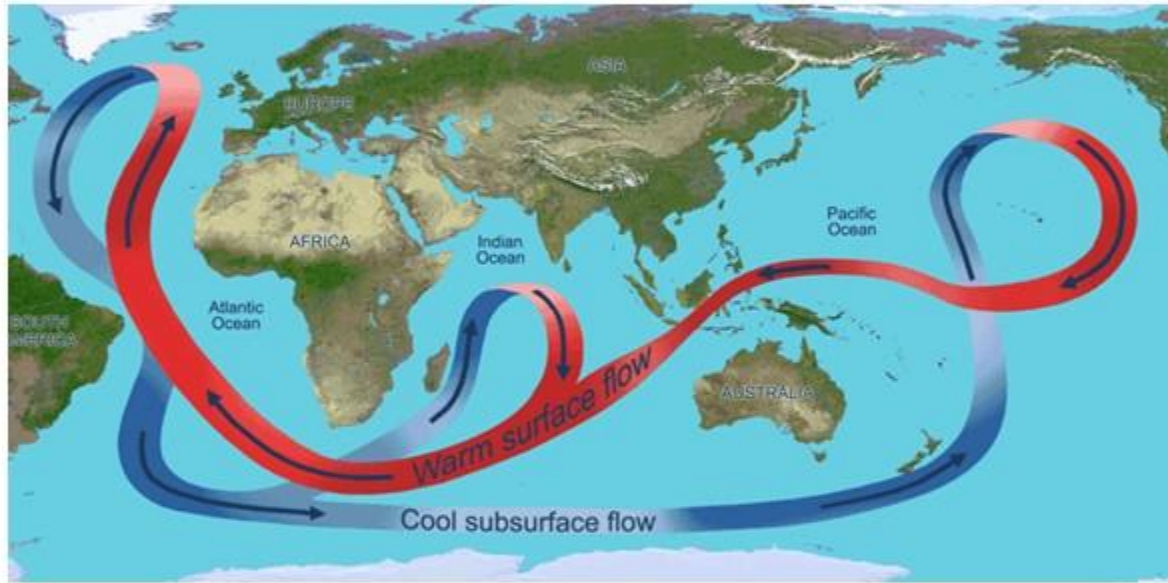
Cold currents bring cold water into warm water areas.

ii. **WARM CURRENTS:**

Warm currents bring warm water into cold water areas.

MOVEMENTS OF OCEAN WATER

MAJOR OCEAN CURRENTS



1 Major ocean currents are greatly influenced by the stresses exerted by the prevailing winds and Coriolis force.

2 Air pressure on the oceanic water causes ocean currents through density variations.

3 The areas of high atmospheric pressure are characterized by low volume of water and thus lowering of water level.

4 Contrary to this, the areas of low atmospheric pressure record higher volume of water and higher water level.

5 Thus, water moves as surface current from the areas of higher water level (low pressure areas) to low water level areas (high pressure areas).

MOVEMENTS OF OCEAN WATER

MAJOR OCEAN CURRENTS



6 In regions of pronounced monsoonal flow, the monsoon winds influence the current movements.

7 Due to the Coriolis force, the warm currents from low latitudes tend to move to the right in the northern hemisphere and to their left in the southern hemisphere.

8 The oceanic circulation transports heat from one latitude belt to another in a manner similar to the heat transported by the general circulation of the atmosphere.

9 The cold waters of the Arctic and Antarctic circles move towards warmer water in tropical and equatorial regions, while the warm waters of the lower latitudes move polewards.

MOVEMENTS OF OCEAN WATER

EFFECTS OF OCEAN CURRENTS

Ocean currents have a number of direct and indirect influences on human activities.

1 DESERT FORMATION

1 Cold ocean currents have a **direct effect on desert formation** in west coast regions of the tropical and subtropical continents.

2 There is **fog** and most of the areas are **arid due to desiccating effect** (loss of moisture).

2 RAINS

1 Warm ocean currents bring rain to coastal areas and even interiors.
Example: Summer Rainfall in British Type climate.



MOVEMENTS OF OCEAN WATER

EFFECTS OF OCEAN CURRENTS

3 MODERATING EFFECT

They are **responsible for moderate temperatures at coasts**. [North Atlantic Drift brings warmness to England. Canary cold current brings cooling effect to Spain, Portugal etc.]

4 DRIZZLE

Mixing of cold and warm ocean currents create **foggy weather** where precipitation occurs in the form of drizzle [Newfoundland].

5 FISHING

1 Mixing of cold and warm ocean currents **bear richest fishing grounds in the world**.

Example: Grand Banks around Newfoundland, Canada and North-Eastern Coast of Japan.

2 The mixing of warm and cold currents help to **replenish the oxygen and favor the growth of planktons**, the primary food for fish population. The best fishing grounds of the world exist mainly in these mixing zones.

MOVEMENTS OF OCEAN WATER

EFFECTS OF OCEAN CURRENTS

6

CLIMATE



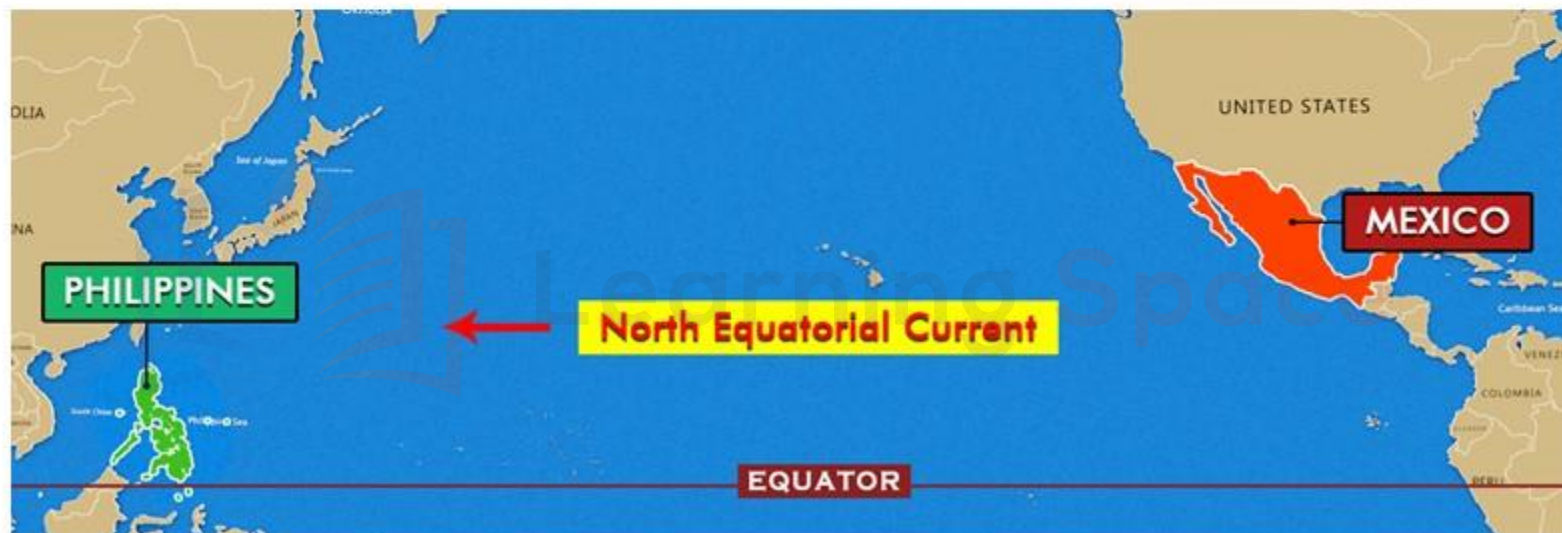
- 1 Warm and rainy climates in tropical and subtropical latitudes [Florida, Natal etc.]
- 2 Cold and dry climates on the western margins in the sub-tropics due to desiccating effect (extreme drying up)
- 3 Foggy weather and drizzle in the mixing zones of cold and warm currents.
- 4 Moderate climate along the western coasts in the sub-tropics.
- 5 Creates Tropical cyclones. They pile up warm waters in tropics and this warm water is the major force behind tropical cyclones.

MOVEMENTS OF OCEAN WATER

EFFECTS OF OCEAN CURRENTS

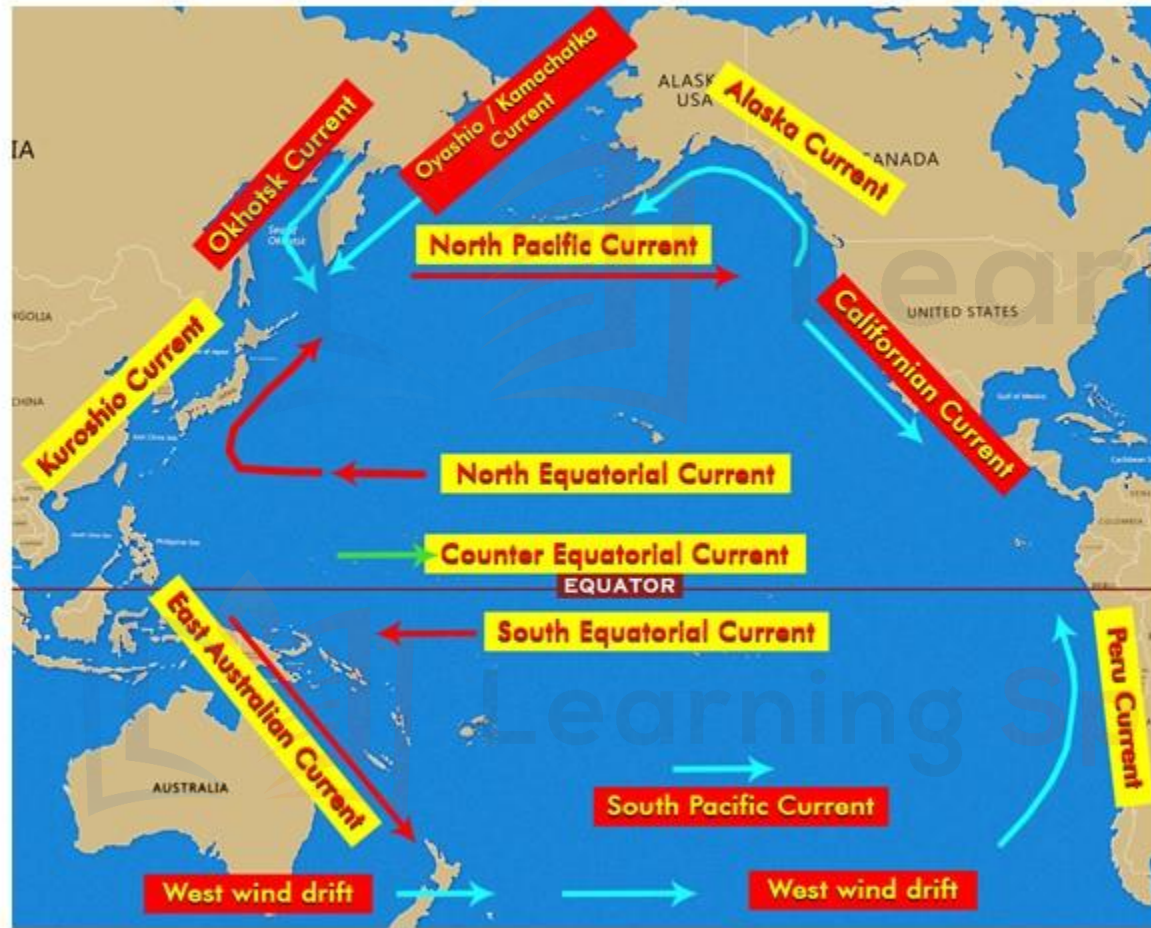
6 NAVIGATION

- 1 Usually, the currents are strongest near the surface and may attain speeds over five knots.
- 2 Ships usually follow routes which are aided by ocean currents and winds.
- 3 For example, if a ship wants to travel from Mexico to Philippines, it can use the route along the North Equatorial Drift which flows from east to west.



MOVEMENTS OF OCEAN WATER

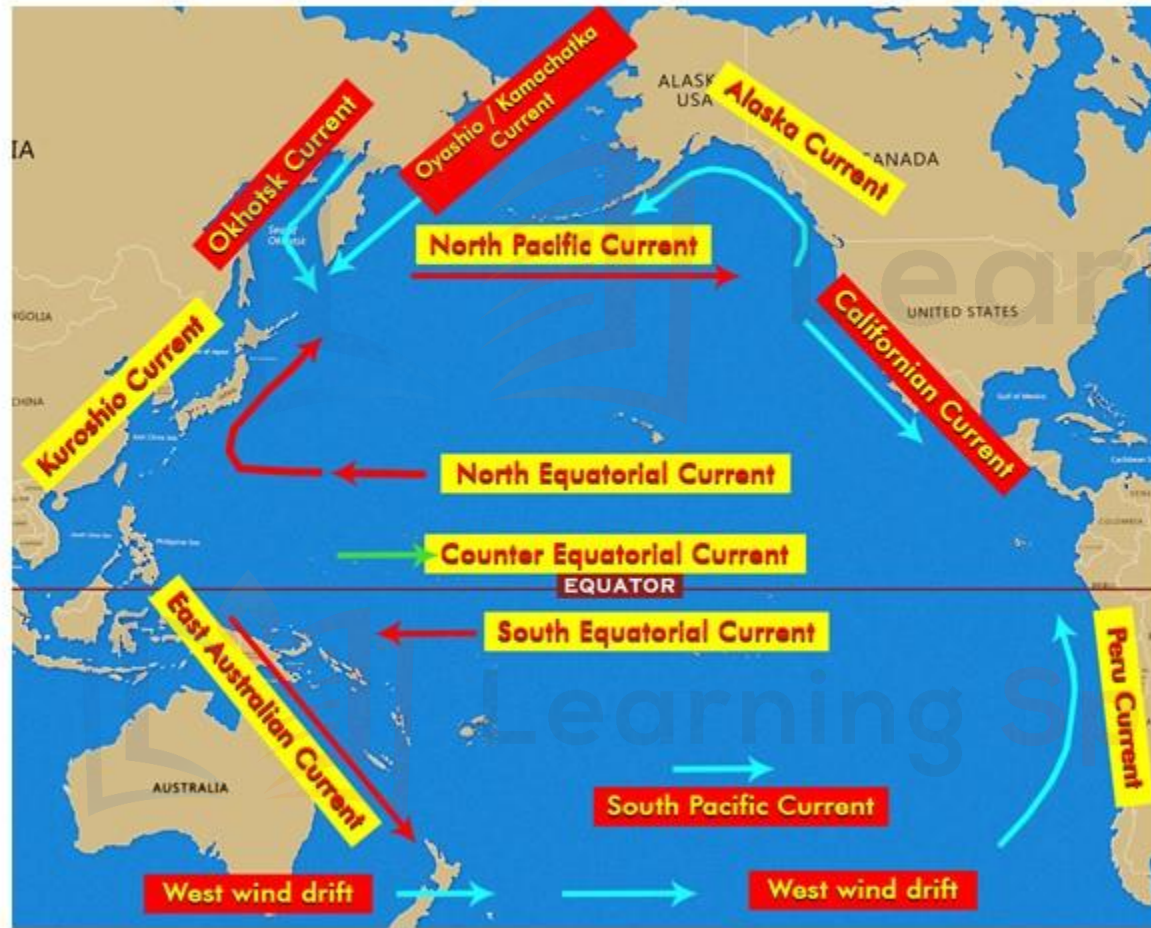
THE CIRCULATION OF THE PACIFIC OCEAN



- 1 The pattern of circulation on the Pacific is similar to that of Atlantic except in modifications which can be expected from the greater size and the more open nature of the Pacific.
- 2 The **North Equatorial Current** flows westwards with a compensating **Equatorial Counter Current** running in the opposite direction.
- 3 Due to the greater expanse of the Pacific and the absence of an obstructing land mass the volume of water is very much greater than that of the Atlantic equatorial current.

MOVEMENTS OF OCEAN WATER

THE CIRCULATION OF THE PACIFIC OCEAN

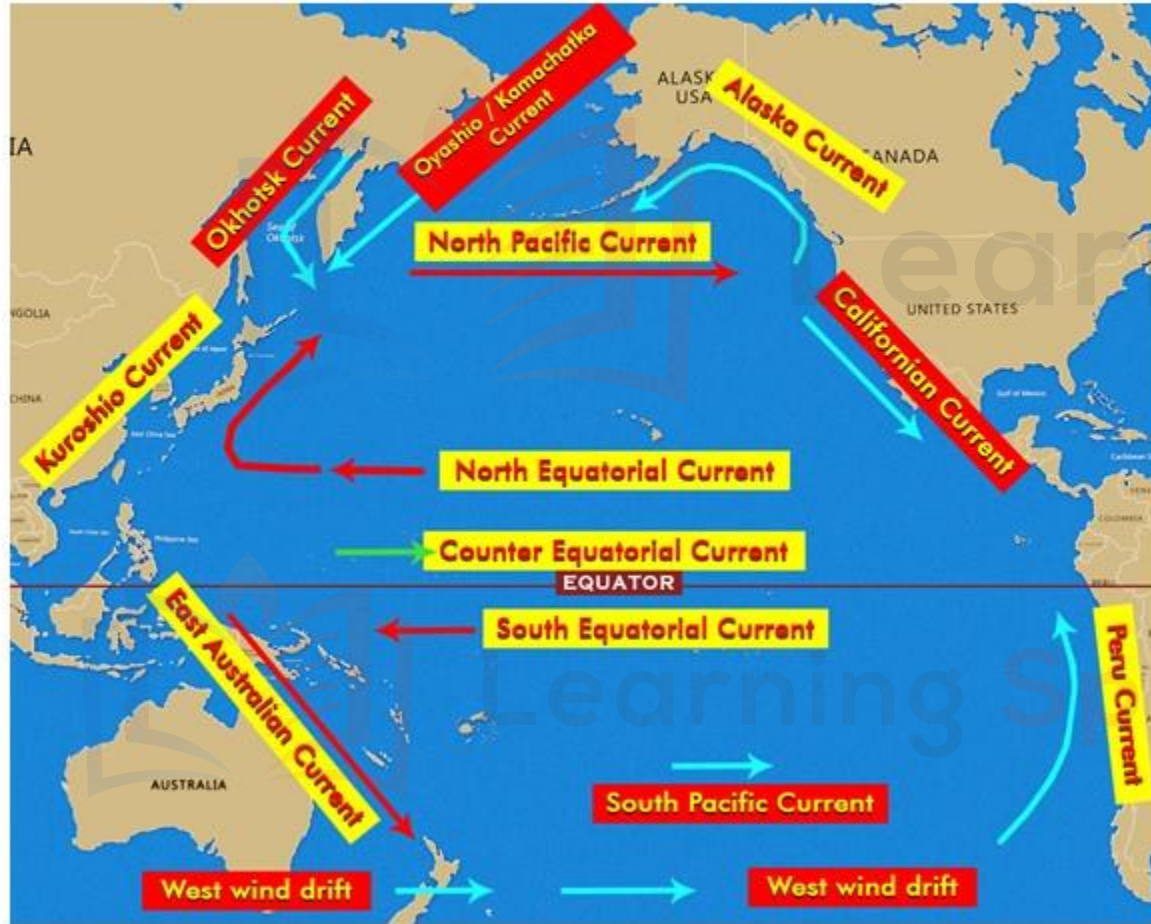


4 The North-East Trade winds blow the North Equatorial Current off the coasts of the Philippines and Formosa into the East China Sea as the **Kuroshio or Japan current**. Its warm waters are carried polewards as the **North Pacific Drift**, keeping the ports of the Alaskan coast ice-free in winter.

5 The cold **Bering Current or Alaskan Current** creeps southwards from the narrow Bering Strait and is joined by the Okhotsk Current to meet the warm Japan Current as the Oyashio, off Hokkaido.

MOVEMENTS OF OCEAN WATER

THE CIRCULATION OF THE PACIFIC OCEAN



- 6 The cold water of **Oyashio** eventually sinks beneath the warmer waters of the North Pacific Drift.
- 7 Part of it drifts eastwards as the cool **California Current** along the coasts of the Western USA and coalesces with the North equatorial Current to complete the clockwise circulation.
- 8 The current system of the South Pacific is the same as that of the South Atlantic.

MOVEMENTS OF OCEAN WATER

THE CIRCULATION OF THE PACIFIC OCEAN



9 The **South Equatorial Current**, driven by the South-East Trade Winds, flows southwards along the coast of Queensland as the **East Australian Current**, bringing warm equatorial waters into temperate waters.

10 The current turns eastward towards New Zealand under the full force of the Westerlies in the Tasman Sea and runs as the **South Pacific current**.

11 Obstructed by tip of the southern Chile, the current turns northwards along the western coast of South America as the **cold Humboldt or Peruvian Current**.

MOVEMENTS OF OCEAN WATER

THE CIRCULATION OF THE PACIFIC OCEAN

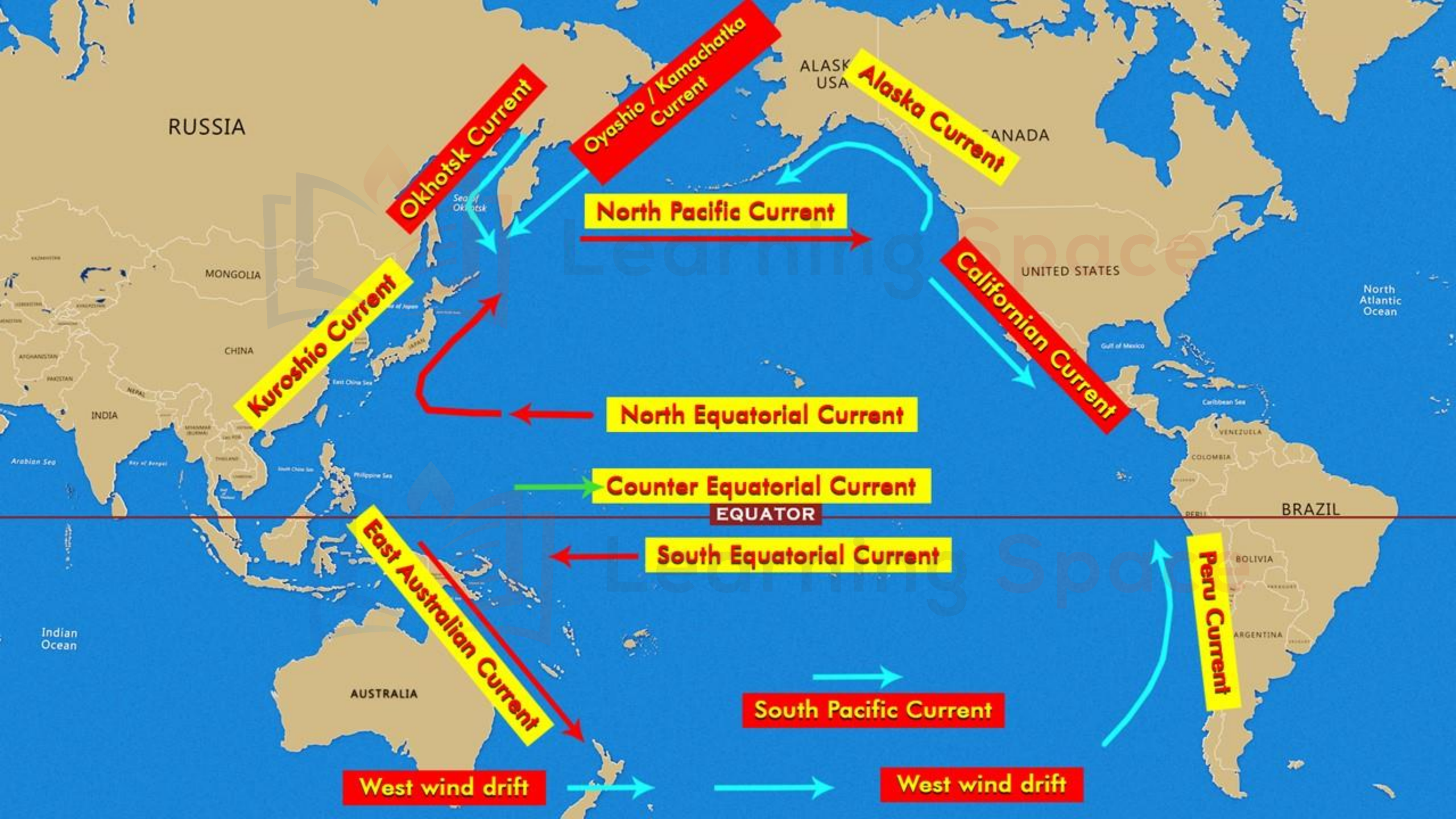
- 12** The cold water influences a desiccating effect so that the **Chilean and Peruvian coasts** are practically rainless.



- 13** The region is **rich in microscopic marine plants and animals** that attract huge shoals of fish. Consequently, **millions of sea-birds gather here** to feed on the fish.

- 14** Their droppings completely whiten the costal cliffs and islands, forming thick deposits of **guano, a valuable source of fertilizer**.

- 15** The Peruvian Current eventually links up with the South Equatorial Current and completes the cycle of currents in the South Pacific.



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